

Week 5 Problem Set

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1. Start a new document, `problemset5.Rmd`, in your `soc10problemsets` repository. (You can download a template here, but be sure to save it in your own `soc10problemsets` repo). Read the North Carolina shapefile into R using the process described in section `@ref(using-simple-features-in-r)`. Then plot the data using `ggplot2`. Use a variable other than `AREA` to fill the shapes, and try using one of the `scale_fill_` arguments to create a red color scale. See these notes on color scales with `ggplot` for some theory and ideas here. **Commit** your changes.

I loaded the North Carolina shapefile using `st_read()` and plotted it using `ggplot()` and `geom_sf()`. Instead of using `AREA` to fill the shapes like the class example, I used `PERIMETER` which shows how the boundary length varies across counties in the state. I used `scale_fill_gradient()` with `pink` and `darkred` to create a red color scale where darker red counties have longer perimeters. I noticed that coastal counties tend to have longer perimeters which makes sense given their irregular coastline boundaries compared to inland counties.

```
library(sf)
```

```
## Warning: package 'sf' was built under R version 4.5.2
```

```
## Linking to GEOS 3.13.0, GDAL 3.8.5, PROJ 9.5.1; sf_use_s2() is TRUE
```

```
library(ggplot2)
```

```
# north carolina file
```

```
nc <- st_read(system.file("shape/nc.shp", package="sf"))
```

```
## Reading layer 'nc' from data source
```

```
##   '/Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/library/sf/shape/nc.shp'
```

```
##   using driver 'ESRI Shapefile'
```

```
## Simple feature collection with 100 features and 14 fields
```

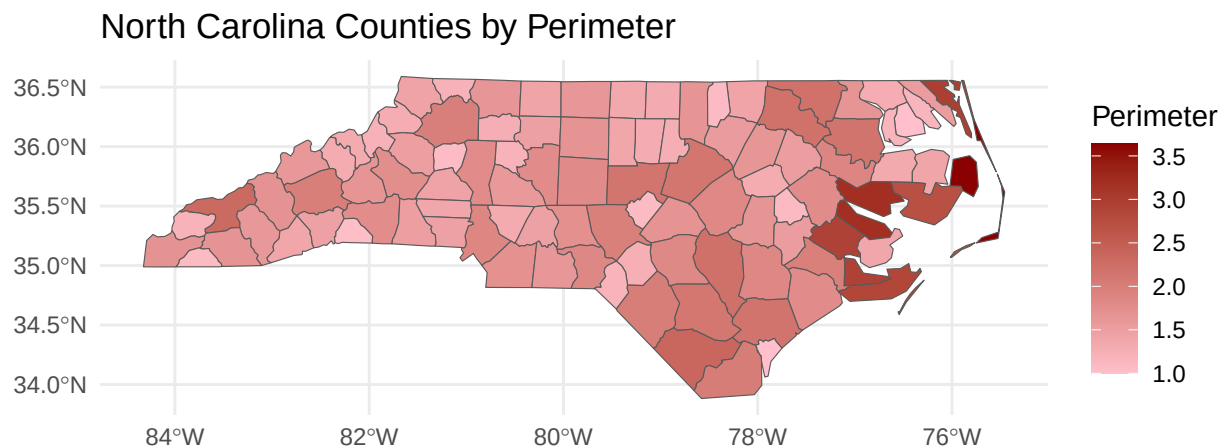
```
## Geometry type: MULTIPOLYGON
```

```
## Dimension:      XY
```

```
## Bounding box:   xmin: -84.32385 ymin: 33.88199 xmax: -75.45698 ymax: 36.58965
```

```
## Geodetic CRS:   NAD27
```

```
ggplot(nc)+  
  geom_sf(aes(fill = PERIMETER))+  
  scale_fill_gradient(low = "pink", high = "darkred")+  
  labs(title = "North Carolina Counties by Perimeter",  
        fill = "Perimeter")+  
  theme_minimal()
```



2. Use the `tidycensus` function to download income data for block groups in Santa Clara County. Plot the demographic data using `ggplot()` and comment on what it shows. **Commit** your changes.

I used `get_acs()` to download median household income data for block groups in Santa Clara County for 2020. Looking at the map, there is a lot of variation in income across the county. The large eastern and southeastern portions of the county show very high incomes in the \$150,000-\$250,000 range shown in the lighter purple and pink tones, which likely corresponds to the wealthier hillside and tech corridor areas. The more densely populated central and western areas around San Jose show much more variation, with darker purple and black areas indicating lower incomes around \$50,000 or below sitting right next to lighter areas with much higher incomes. The gray areas represent block groups where data was not available. This pattern reflects the well known economic inequality in the Bay Area where being close to tech hubs and the high cost of housing create really stark differences in income even across very short distances.

```
library(tidycensus)
```

```
## Warning: package 'tidycensus' was built under R version 4.5.2
```

```
library(ggplot2)
```

```
census_api_key("ffea6c5d2bcd3c8fdf9ff3f6aa412b711e54559")
```

```
# Santa Clara County
```

```
santa_clara <- get_acs(
  state = "CA",
```

```

county = "Santa Clara",
geography = "block group",
variables = "B19013_001",
geometry = TRUE,
year = 2020)

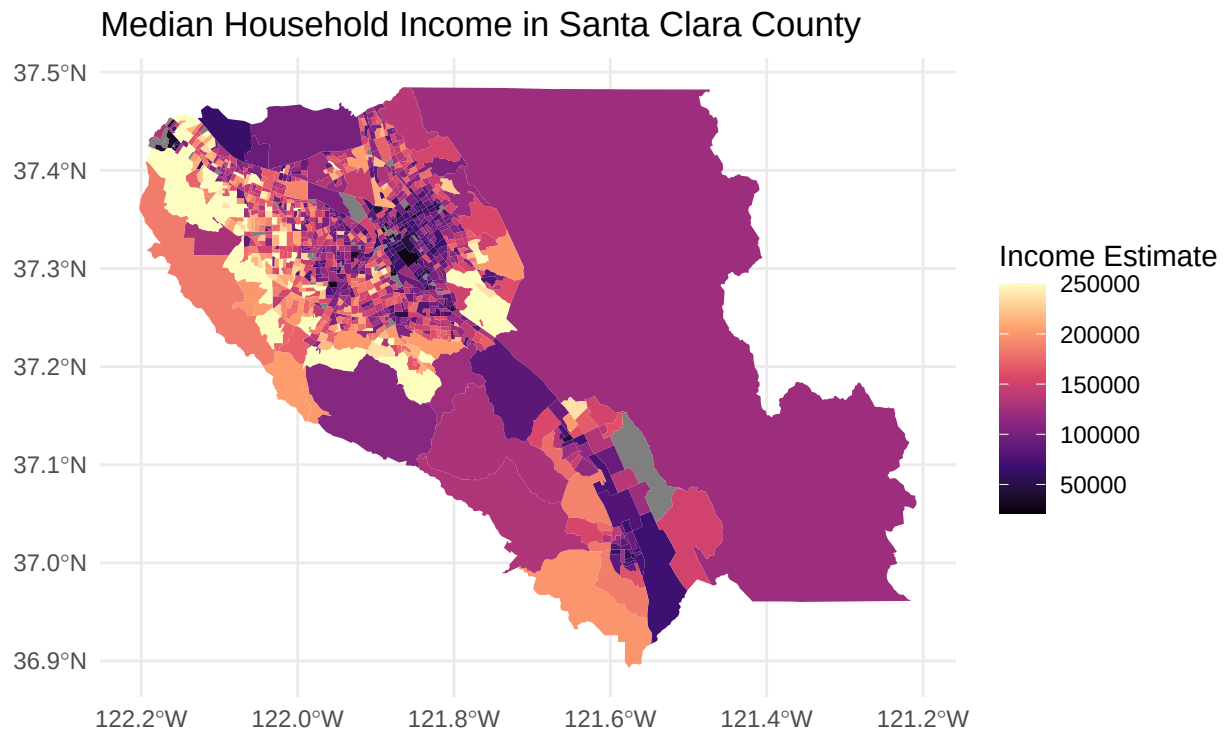
```

```
## |
```

```

# plot
ggplot(data = santa_clara, aes(fill = estimate)) +
  geom_sf(color = NA) +
  scale_fill_viridis_c(option = "magma") +
  labs(title = "Median Household Income in Santa Clara County",
       fill = "Income Estimate") +
  theme_minimal()

```



3. Use the `tigris` function to download spatial data for counties in California. Plot the data using `ggplot()`. **Commit** your changes.

```

library(tigris)
library(ggplot2)

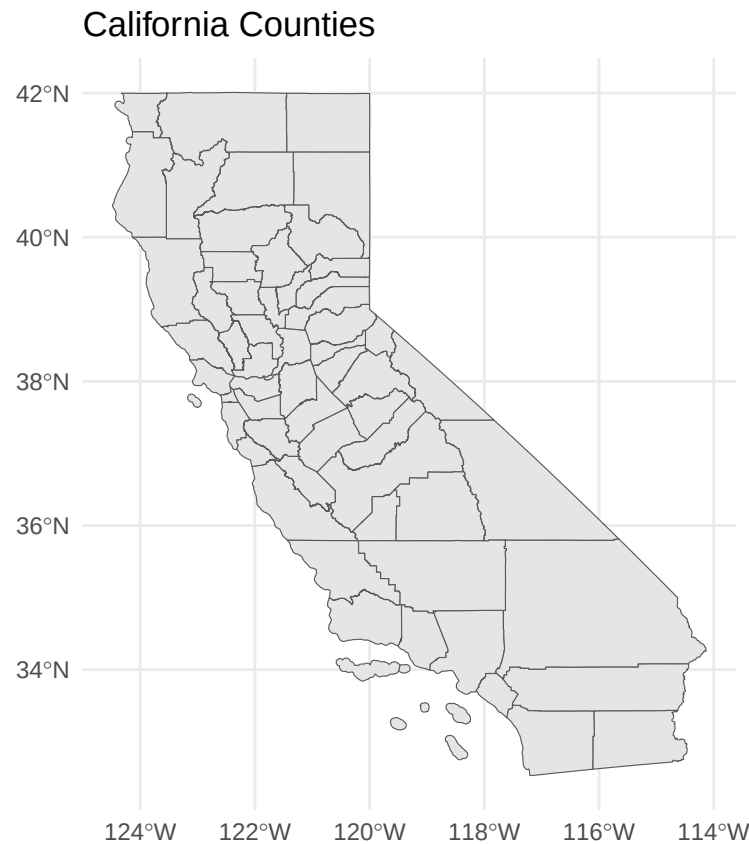
counties <- counties(state = "CA")

```

```
## |
```

```
|
```

```
# plot counties
ggplot(counties)+
  geom_sf()+
  labs(title = "California Counties") +
  theme_minimal()
```



4. Download the clean AQI monitor datasets from Canvas (“aqi22_clean_ca.csv”) or using the code below, and plot these inside the boundaries that you have downloaded from **tigris**. Then, calculate the average PM2.5 levels in each of the geographical units, and plot your geographies again according to these levels (as we did in section @ref(spatially-joining-data)). Which areas had the best (and worst) air quality in 2022? What are the implications? **Commit** your changes.

The first plot shows the PM2.5 monitor locations across California overlaid on county boundaries. The second plot shows the average PM2.5 levels aggregated to the county level. I can see that the Central Valley counties like Fresno had the worst air quality in 2022 while coastal and northern counties had the best. This matters because the Central Valley has a large Latino agricultural and immigrant working population that is lower income, meaning the communities with the worst air quality are also the most economically vulnerable which reflects broader patterns of environmental injustice. Further, medical care is least advanced and accessible for people within these Central Valley regions, further representing how those who face health impacts from bad air quality also will not receive the best medical care and accessibility to receive complete and necessary healthcare.

```
library(readr)
library(RCurl)
```

```
## Warning: package 'RCurl' was built under R version 4.5.2
```

```
library(magrittr)
library(sf)
library(viridis)
library(ggplot2)
library(dplyr)
```

```
## Warning: package 'dplyr' was built under R version 4.5.2
```

```
library(tigris)

# input url from github
url <- getURL("https://raw.githubusercontent.com/tylermcdaniel/css/main/Data/aqi22_clean_ca.csv")

# download aqi monitor data (from github)
aqi22m <- read_csv(url)

aqi22m %<>%
  st_as_sf(coords = c("X", "Y"))

# set crs
aqi22m %<>%
  st_set_crs(st_crs(counties))

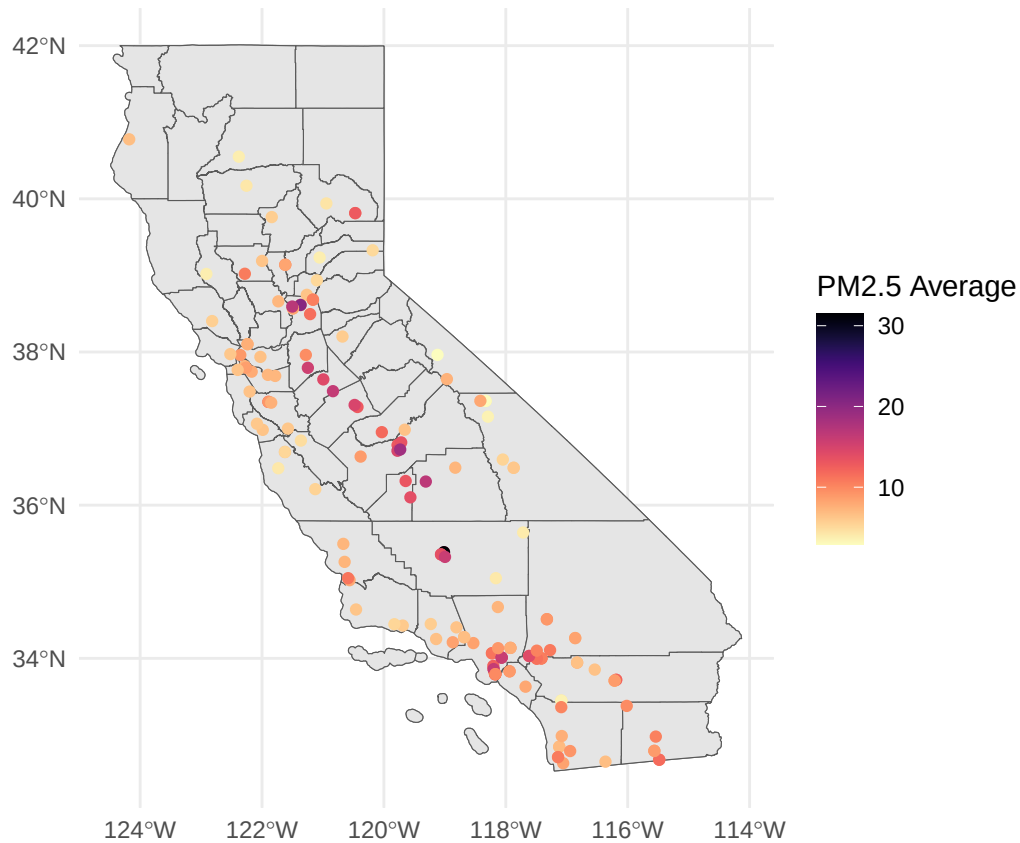
# limit to PM2.5
aqi22m %<>%
  filter(`Pollutant Standard` == "PM25 24-hour 2012" &
         `State Name` == "California")

# cali state boundary
ca <- states() %>%
  filter(NAME == "California")
```

```
## |
```

```
|
```

```
# plot monitors inside county boundaries
ggplot(aqi22m)+
  geom_sf(data = ca)+
  geom_sf(data = counties)+
  geom_sf(aes(col = `Arithmetic Mean`))+
  scale_color_viridis(option = "magma", direction = -1)+
  theme_minimal()+
  labs(col = "PM2.5 Average")
```



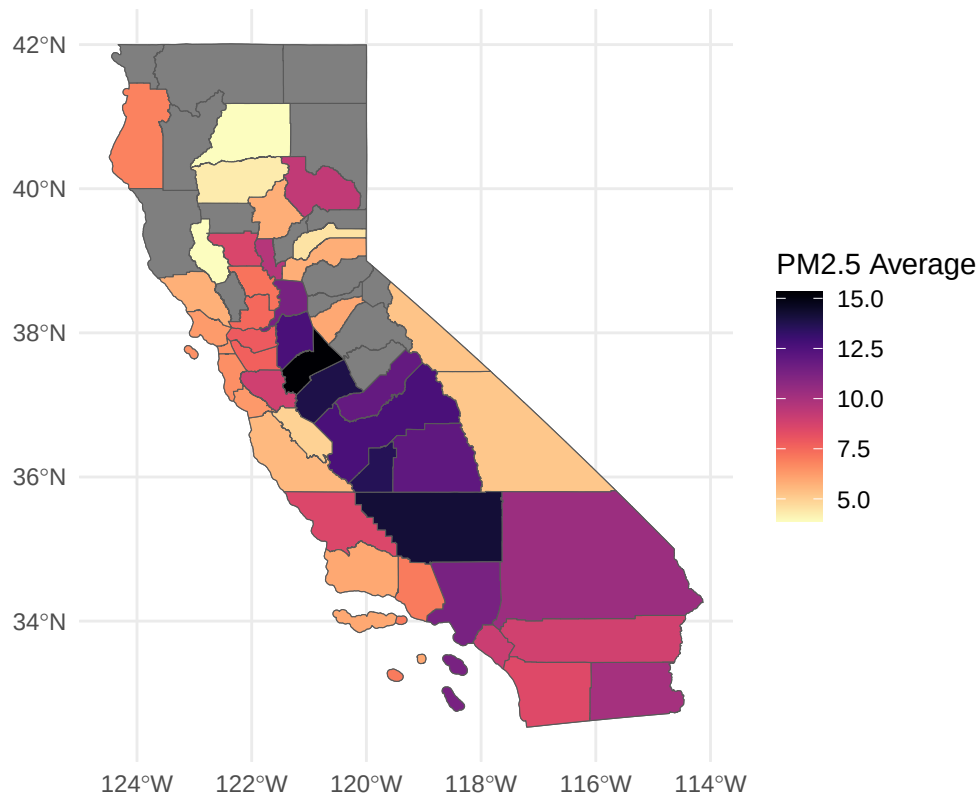
```
# spatial join
counties %<>%
  st_join(aqi22m)

counties %<>%
  group_by(GEOID)

# average AQI for each county
counties %<>%
  summarise(mn_aqi = mean(`Arithmetic Mean`, na.rm = T))

# plot average PM2.5 by county
ggplot(counties)+
  geom_sf(data = ca)+
  geom_sf(aes(fill = mn_aqi))+
  scale_fill_viridis(option = "magma", direction = -1)+
  theme_minimal()+
  labs(fill = "PM2.5 Average",
       title = "Average PM2.5 Levels by County in California 2022")
```

Average PM2.5 Levels by County in California 2022



5. Does the county-level air quality map accurately represent how people experience air quality? Is the Modifiable Areal Unit Problem something that we should be concerned with? **Commit** your changes.

I don't think the county-level air quality map accurately represents how people actually experience air quality. Counties in California are really large and there is a lot of variation within a single county so someone living right next to a highway or industrial facility or oil plant or farm in Fresno experiences very different air quality than someone living on the outskirts of the same county in a more healthy condition. By averaging PM2.5 levels across an entire county we lose that within-county variation which I think could be masking some of the worst pollution exposure that specific communities face so showing the variability is important.

I do think the Modifiable Areal Unit Problem is something we should be concerned with here because the way we draw boundaries can significantly change our findings and if we had used census tracts or block groups instead of counties we would likely see much more dramatic differences in air quality across space, particularly in urban areas where pollution sources like highways and factories tend to be concentrated in specific neighborhoods and not entire cities or counties. Using counties smooths over these differences and could lead us to underestimate how unequally air pollution is distributed. A county like Los Angeles for example contains both very clean and very polluted areas and averaging these together gives a misleading picture of what residents are actually breathing as it varies by neighborhood significantly. I feel that choosing a smaller and more meaningful units like census tracts would give a much more accurate and socially relevant picture of air quality inequality and better reflect the environmental injustice patterns I described in Q4.